`

AL FLASHER 12 V LOW TEMPERATURE OPERATING TEST REPORT

Part Name	AL FLASHER 12 V	**Application**	COMMERCIAL APPLICATION
Customer Part No.	FGG00600	**Customer**	ASHOK LEYLAND
UCAL Part No.	RDEPL/AL/15	**Report No**	R&D/AL/12VF/VAL/26/014

INTRODUCTION:

UCAL is currently supplying 12V AL Flashers to Ashok Leyland for commercial applications. This validation plan has been prepared to define the test requirements and validation activities for the current production parts of the respective product. The testing will be carried out as per the approved validation plan to verify product performance, functional reliability, and compliance with specified requirements under defined test conditions.

PURPOSE:

To evaluate the 12 V flashers assembly after **Low Temperature Operating test.**

TEST SPECIFICATION:

Place the Electronic Component in the constant temperature chamber under room temperature. Connect the power supply device, input and
loading devices outside the chamber so the test specimen can be operated. Lower the temperature gradually when the temperature reaches
-10°C. (As per Drg.) Keep the Electronic Component for 1±0.5 hours under the temperature. Then operate the Electronic Component for 70±2 hours under the temperature. Take the Electronic Component out of the chamber remove the water drops from the surface and keep it for 2
hours at room temperature. Then conduct the performance and functional check of the Electronic Component..

ACCEPTANCE CRITERIA:

All function of the flasher performs as designed before and after the test**.**

RESULT:

SAMPLE NO	PERFORMANCE	APPEARANCE	JUDGEMENT
S:05	O	O	O

O: OK **X: NG** **Δ: To be confirmed**

CONCLUSION:

The performance of the Flasher assembly is acceptable after test. The samples are met the requirements.

Rahul Bongarde **Mr. Rahul Bongarde**	Hussain M **Mr.Hussain Merchant**
Prepared By	**Approved By**

AL FLASHER 12 V LOW TEMPERATURE OPERATING TEST REPORT

DATE:22.04.2026

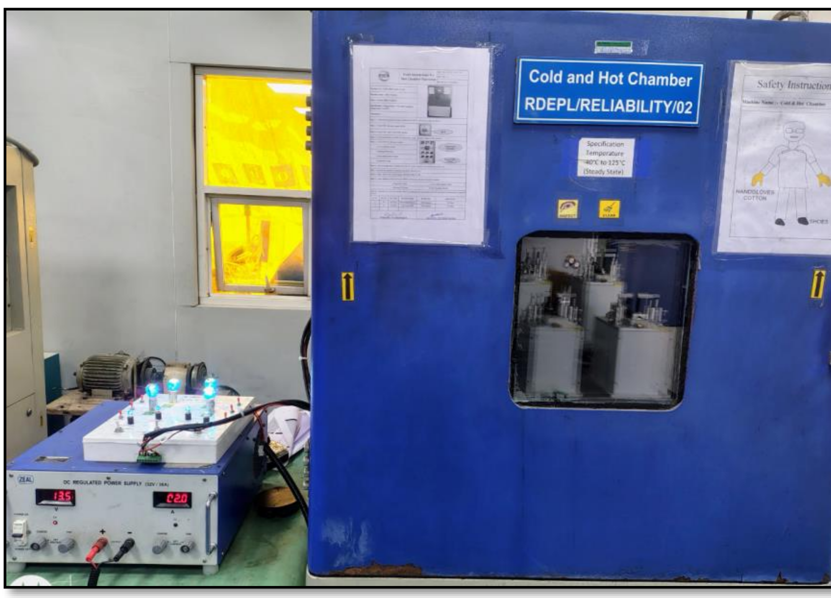
PERFROMACE TEST SPECIFICATIONS:

SNO	PARAMETERS	SPECIFICATIONS
1	Operating voltage	8-16 V
2	Test voltage	13.5 V
3	Flash rate(Turn signal/Hazard warning	60~120 Flashes/Minute
4	One lamp failure flashing rate	>140 Flashes/Min(or)1.8 times of flash rate whichever is greater
5	Duty Cycle	40% ~60%

TEST PROCEDURE:

The test performed with all electrical connections with a lamp load of 2x (2 x 21 W Lamp + 1x 2 W LED) & cluster loads and a voltage of 13.5 V is applied for the duration of 72 Hrs.

TEST SETUP:



TURN SIGNALS & HAZARD SIGNALS:

LEFT SIGNAL	RIGHT SIGNAL	HAZARD SIGNAL
		

AL FLASHER 12 V LOW TEMPERATURE OPERATING TEST REPORT

TEST OBSERVATION:

Description	Load	Observations(at 12 V)	Remark
	LED& CLUSTER	Sample No:01	
Turn Signal RH	2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal	85	OK
Turn Signal LH	2 x 21 W Lamp + 1x2 W LED+ 2 W Cluster Signal	85	OK
Hazard Mode	2x (2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal)	85	OK
Fault condition Any one 21W Lamp failure	2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal	171	OK
Duty cycles for RH&LH	2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal	49%	OK
Duty cycles for Hazard Mode	2x (2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal)	49%	OK

NOTE: The observed flashes are within the specifications & the samples met the performance requirements.

TESTED SAMPLES PHOTOS:

BEFORE TEST SAMPLE – S:05	AFTER TEST SAMPLE – S:05
	

OBSERVATION:

There is no significant change in the electrical performance of the flasher samples. The functionality remains satisfactory. The observed flashes are within the specified requirements & the samples met the performance requirements. All function of the Flasher performs as designed before & after the test.